

Dictionary

It is collection of keys and values.

Properties

```
In [ ]: 1)Mutable
        2)Ordered/Unordered
        3)We can not use indexing and slicing.
        4)Key can not be duplicate
        5)Values can be duplicates.
        6)Enclosed by {}
        7)Initialize dictionary by using dict() or {}

Python2->Unordered
Python 3->Ordered

Data Types of key:-Immutable(int,float,complex,str,tuple,)

Data Types of value:-Any Data type(int,float,complex,str,list,tuple,set,dict)

dict1=
```

Syntax:

```
In [ ]: dict1={key:Value}

dict2={}=dict()  #dictionary initialization
```

```
In [2]: d1={"a":97,"b":98,"c":122}
        d1
```

```
Out[2]: {'a': 97, 'b': 98, 'c': 122}
```

```
In [3]: d1={"a":97,"b":98,"c":122}
        d1["a"]
```

```
Out[3]: 97
```

```
In [4]: d1={"a":97,"b":98,"c":122}
        d1["B"]
```

KeyError

Traceback (most recent call last)

Cell **In[4]**, line 2

```
1 d1={"a":97,"b":98,"c":122}
----> 2 d1["B"]
```

KeyError: 'B'

```
In [5]: d2={"a":33,"b":22,"c":[111,222,333]}
d2
```

```
Out[5]: {'a': 33, 'b': 22, 'c': [111, 222, 333]}
```

```
In [6]: d2={"a":33,"b":22,"c">{111,222,333}}
d2
```

```
Out[6]: {'a': 33, 'b': 22, 'c': {111, 222, 333}}
```

```
In [7]: len(d2)
```

```
Out[7]: 3
```

Dictionary Functions

1.get()

```
In [ ]: dict.get("key")
```

```
In [9]: dict1={'A':100,'B':200,'C':300,'D':400}
dict1['A']
```

```
Out[9]: 100
```

```
In [11]: dict1={'A':100,'B':200,'C':300,'D':400}
dict1.get('A')
```

```
Out[11]: 100
```

2)keys()

```
In [ ]: dict.keys()
```

```
In [12]: dict1={'A':100,'B':200,'C':300,'D':400}
dict1.keys()
```

```
Out[12]: dict_keys(['A', 'B', 'C', 'D'])
```

```
In [13]: dict1={'A':100,'B':200,'C':300,'D':400}
for i in dict1.keys():
    print(i)
```

```
A
B
C
D
```

```
In [15]: employee_data={'Name':'Yash','Age':22,'Designation':"Data Scientist",'Salary':70000}
len(employee_data)
```

Out[15]: 4

```
In [16]: employee_data.get('Designation')
```

Out[16]: 'Data Scientist'

```
In [17]: employee_data.keys()
```

Out[17]: dict_keys(['Name', 'Age', 'Designation', 'Salary'])

```
In [20]: employee_data={'Name':'Yash','Age':22,'Designation':"Data Scientist",'Salary':70000}
for key in employee_data.keys():
    print("key is:",key)
    print("Values is",employee_data[key])
    print('*'*40)
```

```
key is: Name
Values is Yash
*****
key is: Age
Values is 22
*****
key is: Designation
Values is Data Scientist
*****
key is: Salary
Values is 70000
*****
```

3)values()

```
In [ ]: Syntax:Dict.values()
```

```
In [21]: employee_data={'Name':'Yash','Age':22,'Designation':"Data Scientist",'Salary':70000}
employee_data.values()
```

Out[21]: dict_values(['Yash', 22, 'Data Scientist', 70000])

```
In [22]: employee_data={'Name':'Yash','Age':22,'Designation':"Data Scientist",'Salary':70000}
for value in employee_data.values():
    print("Values is:",value)
    print('*'*40)
```

```
Values is: Yash
*****
Values is: 22
*****
Values is: Data Scientist
*****
Values is: 70000
*****
```

4)items()

In []: Syntax: Dict.items()

In [23]: `employee_data={'Name':'Yash','Age':22,'Designation':"Data Scientist",'Salary':70000}`
`employee_data.items()`

Out[23]: dict_items([('Name', 'Yash'), ('Age', 22), ('Designation', 'Data Scientist'), ('Salary', 70000)])

In [24]: `employee_data={'Name':'Yash','Age':22,'Designation':"Data Scientist",'Salary':70000}`
`for i in employee_data.items():`
 `print(i)`

('Name', 'Yash')
 ('Age', 22)
 ('Designation', 'Data Scientist')
 ('Salary', 70000)

In [26]: `a,b=('Name', 'Yash')`
`a`

Out[26]: 'Name'

In [27]: `b`

Out[27]: 'Yash'

In [28]: `employee_data={'Name':'Yash','Age':22,'Designation':"Data Scientist",'Salary':70000}`
`for key,value in employee_data.items():`
 `print(f"key=={key} and value=={value}")`
 `print("*"*40)`

key==Name and value==Yash

 key==Age and value==22

 key==Designation and value==Data Scientist

 key==Salary and value==70000

5)Update

In []: Update method Inserts the specified items to the dictionary

In [1]: `employee_data={'Name':'Yash','Age':22,'Designation':"Data Scientist",'Salary':70000}`
`employee_data`

Out[1]: {'Name': 'Yash', 'Age': 22, 'Designation': 'Data Scientist', 'Salary': 70000}

```
In [2]: employee_data['Age']
```

```
Out[2]: 22
```

```
In [3]: employee_data['Age']=32  
employee_data
```

```
Out[3]: {'Name': 'Yash', 'Age': 32, 'Designation': 'Data Scientist', 'Salary': 7000  
0}
```

```
In [5]: employee_data.update({"Salary":80000})  
employee_data
```

```
Out[5]: {'Name': 'Yash', 'Age': 32, 'Designation': 'Data Scientist', 'Salary': 8000  
0}
```

```
In [7]: employee_data={'Name':'Yash','Age':22,'Designation':"Data Scientist",'Salary':  
employee_data.update({"Location":"Pune"})  
employee_data
```

```
Out[7]: {'Name': 'Yash',  
        'Age': 22,  
        'Designation': 'Data Scientist',  
        'Salary': 70000,  
        'Location': 'Pune'}
```

```
In [10]: list1=[2,3,4,5,6,7,8]  
#square_dict1={2:4,3:9,4:16,5:25,6:36,7:49,8:64}  
square_dict1={}  
for i in list1:  
    square=i**2  
    square_dict1.update({i:square})  
print(square_dict1)
```

```
{2: 4, 3: 9, 4: 16, 5: 25, 6: 36, 7: 49, 8: 64}
```

```
In [11]: A={}  
A.update({'A':100})  
A.update({'B':200})  
A
```

```
Out[11]: {'A': 100, 'B': 200}
```

```
In [12]: square=dict()
         for i in range(11,21):
             s=i**2
             square.update({i:s})
         square
```

```
Out[12]: {11: 121,
          12: 144,
          13: 169,
          14: 196,
          15: 225,
          16: 256,
          17: 289,
          18: 324,
          19: 361,
          20: 400}
```

ASCII

```
In [ ]: {'A':65,'B':66.....'Z':90}
```

```
In [16]: ascii={}
         for i in range(65,91):
             #print(i,chr(i))
             ascii.update({chr(i):i})
         ascii
```

```
Out[16]: {'A': 65,
          'B': 66,
          'C': 67,
          'D': 68,
          'E': 69,
          'F': 70,
          'G': 71,
          'H': 72,
          'I': 73,
          'J': 74,
          'K': 75,
          'L': 76,
          'M': 77,
          'N': 78,
          'O': 79,
          'P': 80,
          'Q': 81,
          'R': 82,
          'S': 83,
          'T': 84,
          'U': 85,
          'V': 86,
          'W': 87,
          'X': 88,
          'Y': 89,
          'Z': 90}
```

Delete Items From Dictionary

```
In [ ]: 1)pop()  
        2)popitem  
        3)clear
```

1)pop()

```
In [ ]: Dict.pop(key)
```

```
In [17]: data={'A':100,'B':200,'C':300,'D':400,'E':500}  
         data.pop('B')  
         data
```

```
Out[17]: {'A': 100, 'C': 300, 'D': 400, 'E': 500}
```

```
In [19]: data={'A':100,'B':200,'C':300,'D':400,'E':500}  
         data.pop('B')  
         data.pop('A')  
         data
```

```
Out[19]: {'C': 300, 'D': 400, 'E': 500}
```

2)popitem()

```
In [ ]: Dict.popitem()
```

It will delete last inserted item (key **and** value pair)

```
In [20]: data={'A':100,'B':200,'C':300,'D':400,'E':500}  
         data.popitem()  
         data
```

```
Out[20]: {'A': 100, 'B': 200, 'C': 300, 'D': 400}
```

```
In [21]: data={'A':100,'B':200,'C':300,'D':400,'E':500}  
         data.popitem()  
         data.popitem()  
         data
```

```
Out[21]: {'A': 100, 'B': 200, 'C': 300}
```

3)clear()

```
In [ ]: Dict.clear()
```

It will delete **all** items **from** dictionary

It **return** Null dictionary.

```
In [22]: data={'A':100,'B':200,'C':300,'D':400,'E':500}
data.clear()
data
```

Out[22]: {}

```
In [23]: data
```

Out[23]: {}

4. Del keyword

```
In [24]: data={'A':100,'B':200,'C':300,'D':400,'E':500}
data['C']
```

Out[24]: 300

```
In [27]: data['C']=350
data
```

Out[27]: {'A': 100, 'B': 200, 'C': 350, 'D': 400, 'E': 500}

```
In [29]: del data['C']
data
```

```
-----
KeyError                                Traceback (most recent call last)
Cell In[29], line 1
----> 1 del data['C']
      2 data

KeyError: 'C'
```

```
In [31]: data={'A':100,'B':200,'C':300,'D':400,'E':500}
print(data)
del data
print(data)
```

{'A': 100, 'B': 200, 'C': 300, 'D': 400, 'E': 500}

```
-----
NameError                                Traceback (most recent call last)
Cell In[31], line 4
      2 print(data)
      3 del data
----> 4 print(data)

NameError: name 'data' is not defined
```

fromkeys

```
Dict.fromkeys(iterable,[Value])
```

```
Create dictionary with same value.
```

```
Default value is None.
```

```
In [32]: x=list("ASDFG")  
x
```

```
Out[32]: ['A', 'S', 'D', 'F', 'G']
```

```
In [33]: x=list("Python")  
x
```

```
Out[33]: ['P', 'y', 't', 'h', 'o', 'n']
```

```
In [34]: new_dict=dict.fromkeys(x)  
new_dict
```

```
Out[34]: {'P': None, 'y': None, 't': None, 'h': None, 'o': None, 'n': None}
```

```
In [35]: new_dict=dict.fromkeys('Python',500)  
new_dict
```

```
Out[35]: {'P': 500, 'y': 500, 't': 500, 'h': 500, 'o': 500, 'n': 500}
```

```
In [36]: new_dict=dict.fromkeys((2,3,4,5,6),500)  
new_dict
```

```
Out[36]: {2: 500, 3: 500, 4: 500, 5: 500, 6: 500}
```

```
In [37]: new_dict=dict.fromkeys('Python',[10,20,30,40,50])  
new_dict
```

```
Out[37]: {'P': [10, 20, 30, 40, 50],  
          'y': [10, 20, 30, 40, 50],  
          't': [10, 20, 30, 40, 50],  
          'h': [10, 20, 30, 40, 50],  
          'o': [10, 20, 30, 40, 50],  
          'n': [10, 20, 30, 40, 50]}
```

```
In [38]: new_dict=dict.fromkeys('Python','Data science')  
new_dict
```

```
Out[38]: {'P': 'Data science',  
          'y': 'Data science',  
          't': 'Data science',  
          'h': 'Data science',  
          'o': 'Data science',  
          'n': 'Data science'}
```

```
In [42]: sub_name=["Bio", "Chem", "Hindi", "Math", "Eng", 'Phy']
marks=[88,90,76,65,87,80]
result={}
for i,value in enumerate(sub_name):
    result.update({value:marks[i]})
print(result)
```

```
{'Bio': 88}
{'Bio': 88, 'Chem': 90}
{'Bio': 88, 'Chem': 90, 'Hindi': 76}
{'Bio': 88, 'Chem': 90, 'Hindi': 76, 'Math': 65}
{'Bio': 88, 'Chem': 90, 'Hindi': 76, 'Math': 65, 'Eng': 87}
{'Bio': 88, 'Chem': 90, 'Hindi': 76, 'Math': 65, 'Eng': 87, 'Phy': 80}
```

setdefault()

```
In [ ]: Dict.setdefault(key,[Val])
```

Default value is **None**.

```
In [43]: result={'Bio': 88, 'Chem': 90, 'Hindi': 76, 'Math': 65, 'Eng': 87, 'Phy': 80}
result['Chem']
```

Out[43]: 90

```
In [44]: result.get('Chem')
```

Out[44]: 90

```
In [46]: result={'Bio': 88, 'Chem': 90, 'Hindi': 76, 'Math': 65, 'Eng': 87, 'Phy': 80}
result.setdefault('Chem')
result
```

Out[46]: {'Bio': 88, 'Chem': 90, 'Hindi': 76, 'Math': 65, 'Eng': 87, 'Phy': 80}

```
In [50]: result={'Bio': 88, 'Chem': 90, 'Hindi': 76, 'Math': 65, 'Eng': 87, 'Phy': 80}
result.update({'Chem':100})
result
```

Out[50]: {'Bio': 88, 'Chem': 100, 'Hindi': 76, 'Math': 65, 'Eng': 87, 'Phy': 80}

```
In [51]: result.setdefault("Chem",67)
result
```

Out[51]: {'Bio': 88, 'Chem': 100, 'Hindi': 76, 'Math': 65, 'Eng': 87, 'Phy': 80}

```
In [52]: result={'Bio': 88, 'Chem': 90, 'Hindi': 76, 'Math': 65, 'Eng': 87, 'Phy': 80}
result.setdefault('Python')
result
```

```
Out[52]: {'Bio': 88,
          'Chem': 90,
          'Hindi': 76,
          'Math': 65,
          'Eng': 87,
          'Phy': 80,
          'Python': None}
```

```
In [ ]:
```